

Stage 11 Revision 45 Dependency, Force, and Transition Consistency

mdstats

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Purpose

This specification prevents recurrence of the revision-44 stage-order, ensemble-force, cross-fitting, and transition-promotion contradictions.

Required contracts

- The authoritative dependency graph is revision 45 and is a DAG with product-specific gates.
- Control-inferred dynamics, realized trajectory consistency, and thermodynamic admissibility are separate records.
- E3A mechanical refinement is distinct from E3B canonical thermodynamic mean-force validation.
- Canonical force-density and harmonic equipartition identities are not applied generally to NVE.
- EvidenceCrossfitPartition contains discovery, model selection, basin validation, corridor validation, thermodynamic validation, and optional final-refit domains.
- SAMP2 stops at preliminary corridor and saddle-candidate support.
- E6 owns FinalTransitionEventCertificate.
- THERMO3A owns static finite-region thermodynamics; THERMO3B owns event/path-conditioned transition-state validation and any validated_transition_saddle promotion.
- THERMO4A precedes E8b thermodynamic comparison; THERMO4B follows empirical/barrier rates.
- Stage 11F0 owns RateBoundModel; Stages 11G–11I have explicit roadmap deliverables.
- Historical implementation bullets belong to the status appendix, not the normative manual.